Study Guide: Schedule & Activities Overview

Dear Participant,
Thank you for taking part in our study on evaluating Requirements Engineering (RE) practices for capturing explainability requirements in AI-based systems. Below is the schedule that outlines how the session will be conducted:

Study Organization Overview

- **Total Number of Participants:** 9
- **Study Phases:**
  - The session will pass through **3 phases**
- **Grouping:**
  - Participants will be divided into **3 groups**
  - Each group will include **3 participants**
  - The group members will be reassigned for each of the 3 phases
- **Practice Assignment:**
  - Each group will be assigned **one RE practice per phase** (elicitation, specification, validation)
  - Different groups will use **different practices** within each phase
- **Study Preparation:**
  - Before starting the tasks, **all participants will be provided with:**
    - A brief **scenario** relevant to the system under investigation (e.g., Active Side Guard Assist)
    - **Worked-out examples** for each RE practice to guide their understanding
- **Execution:**
  - Participants will apply their assigned practice using the provided **templates and instructions.**
  - They will follow the **think-aloud protocol**, verbalizing thoughts as they perform tasks.
  - Further, paper and pen will also be provided if it is difficult for participants to verbalize their thoughts.
- **Post-task:**
  - After each phase, **groups will gather again and discuss outcomes and challenges in**
  - All gathered requirements by the three groups will be merged into a unified set of requirements and serve as an input to the next phase.
  - The participants will be shuffled after each phase to mitigate the team confounding effect.

Confidentiality and Anonymity

a. All information provided by the participant shall remain confidential and will not be disclosed to anyone outside the research team.
b. All results will be anonymized before publication.
c. Complete transcripts will NOT be published.
d. Study data will be analysed and where needed direct quotes will be used. They will be attributed to a participant ID, not a clearly identifiable individual.
e. We will ask for explicit usage permission for each direct quote from the respective participant.

- **Study Execution and Analysis**
  An introduction about the XAI and RE will be provided to the participants so that they are familiar with it and can gather information beforehand should they decide to do so. The study time is estimated between 3 - 3.5 Hrs. During the study, we will ask questions about the topic. We would like to take notes and record the process (as participants verbalized their thoughts) with your permission. The recording will be solely used to help us, so we might not miss important information. Thereafter, we will send back the final analyzed data for their approval.

📅 Study Schedule

Time	Activity
8:30 – 8:50 AM	📢 **Introductory Session & Presentation** Welcome, study objectives, overview of RE practices, explanation of the think-aloud protocol.
8:50 – 9:20 AM	✍️ **Phase 1: Applying Elicitation Practices** You will apply your assigned elicitation technique (interview, BS, or task analysis) using the provided templates and system scenario.
	Break
9:30 – 10:00 AM	💬 **Discussion on Elicitation Outcomes** Each group will share what requirements they gathered and reflect on challenges or gaps encountered. All collected requirements will result in a merged set of requirements. The groups will be shuffled for the next session.
10:00 – 10:30 AM	📏 **Phase 2: Applying Specification/Modelling Practices** Using the common requirement set, you will apply your assigned specification technique (natural language, use case, or soft goal modelling).
	Break
10:40 – 11:10 AM	💬 **Discussion on Specification Results** Participants will discuss the clarity, completeness, and challenges that they experienced while specifying explainability requirements. All specified requirements will result in a merged set of requirements. The groups will again be shuffled for the next session.
11:10 – 11:40 AM	☑️ **Phase 3: Applying Validation Practices** Each group will use a different validation method (feedback session, walkthrough, or compliance check) to assess the specified requirements.
11:40 – 12:00 PM	🎯 **Wrap-Up Session & Feedback** We will summarize the session, gather your reflections, and answer any final questions.

📌 Important Notes

- No prior experience in software engineering is required. You'll receive clear templates and guidance for each task.
- Please actively verbalize your thoughts during tasks (think-aloud protocol). We will also provide you sheets if you want to note it down or feel it difficult to verbalize.
- Your insights are valuable in improving how we gather and use explainability requirements in real-world AI systems.